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**EE-8303 (GS)**  
**B.E. VIII Semester** Examination, June 2020  
**Grading System (GS)**  
**Computer Applications to Power Systems**  
(Elective - III)  
**Time : Three Hours**

**Maximum Marks : 70**

**Note :** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. Discuss in detail about the generation and absorption of reactive power in power system components.  
OR  
Explain with diagram, the operation of a fixed capacitor and thyristor controlled reactor.
2. Describe how sensitivity relations predict changes in reactive power generation with changes in PV bus voltage for anticipatory preventive control.
3. What is capability curve of a alternator? Describe the importance of this curve with a neat sketch in power system.
4. What are the main components of a power system? Explain the modeling of any one of the component.
5. Explain the following:
  - i) Generation shift distribution factors
  - ii) Compensated shift factors
6. With neat sketches, explain the following:
  - i) Oriented graph
  - ii) Tree
  - iii) Singular matrix
  - iv) Basic cut sets
  - v) Basic loops
7. Why series compensation is preferred over shunt compensation for long transmission line. What are the demerits of series compensation?
8. Explain the following (Any Two):
  - i) Line loadability
  - ii) Regulating Transform
  - iii) Corrective rescheduling
  - iv) Effect of load module
  - v) Participation factor

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